Hyporheic Flopy Automation

1. Project Overview

- **Purpose**: Automate the setup, execution, and analysis of MODFLOW/MODPATH models for hyporheic exchange studies using FloPy.
- **Workflow**: Preprocess Input Files → Initialize Model → Define Model Domain → Define Boundary Conditions → Run simulations → Analyze and Export Results.

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2. Detailed Workflow

2.1 Preprocessing of Input Layers

- Import and process input rasters/vectors (e.g., DEM, land cover, hydrography).
- Reproject, resample, and clip layers to model domain.
- Validate and fill missing data.

2.2 Model Initialization

- Create model workspace and input directory structure.
- Initialize FloPy MODFLOW model object.
- Define/discretize grid (DIS package): dimensions, cell sizes, top/bottom arrays.
  - rows
  - columns
  - layers
  - depth of model
  - resolution
- Initialize additional model packages as needed:
  - Hydraulic properties (LPF/NPF)
  - Recharge (RCH)
  - River (RIV/CHD)
  - Well (WEL), Drain (DRN), etc.

2.3 Model Domain and Boundary Conditions

- Define model domain
  - Top elevations
  - Grid
- Set up active/inactive cells (IBOUND, idomain).
- Assign boundary conditions spatially (e.g., river, recharge zones, constant head).

- left and right flood plain
- upstream and downstream elevations
- Assign initial conditions (STR, SFR, GHB, or custom packages).
- Visualize domain and BC setup.

2.4 Model Execution

2.4.1 Groundwater Flow (GWF)

- Feed processed input data into GWF model
- Run GWF model simualtion
- Write input/output files and run MODFLOW 6 simulation via FloPy.

2.4.2 Particle Tracking (MODPATH)

- Set up MODPATH model (input/output control, release points).
- Run forward and backward particle tracking using MODPATH 7 for pathline and endpoint tracking.
- Write input/output pathline/endpoints for analysis.

2.5 Post-processing & Results Analysis

- Read and Visualize MODFLOW results: 3D model domain, gwf heads, and surface elevation
- Read MODPATH results (pathlines, endpoints).
- Convert particle locations (x, y, z) to model indices (row, col, layer).
- Calculate per-segment and per-particle statistics:
    - Residence time, path length, velocity, flow rate
    - Extract heads, gradients, and other cell properties
    - Hyporheic volume per cell and along pathlines
- Calculate Zone Budget of Pathlines
- Generate summary statistics (mean, min, max, range for each variable).

2.6 Exporting Results

- Export output table of summary statistics
- Export points and pathlines as GeoDataFrames.
- Write shapefiles/GeoPackages for GIS visualization:
    - Particle pathlines (LineString)
    - Cell/passage points (Point)
- Export summary tables (CSV) for per-cell and per-particle metrics.
- Generate 3D and map plots (matplotlib) for domain, pathlines, and variables.
- Export ESRI-compatible tables and files.

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3. Extensibility & Configuration

- Modularized code for each workflow step (preprocessing, model initialization, model domain, set boundary conditions, run MODFLOW 6 and MODPATH 7 simulations, postprocessing, export).
- User-configurable parameters.
- Batch Processes
- Follow-along public HTML
- replicatable HTML for users
- Easily swap out input data or adjust domain settings.

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4. How to Use

1. Edit Batch File "run_models.bat" to direct inputs to user-specified sources.
2. Type ".\run_models.bat" into terminal to run automated workflow
3. Open results from "Hyporheic_output" or user-specified output folder

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5. References

- FloPy Documentation
- MODFLOW Documentation
- MODPATH Documentation